Supporting Information: Supporting Information for "Calibrating the Interpretation of Docking-Derived Resistance-Retention Scores with a Short Molecular-Dynamics Structural Stress Test from African Natural Products"

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State-specific summaries. For PfDHFR, mean RRS values were 74.7 (N51I), 73.7 (C59R), 75.3 (S108N), and 76.8 (I164L). For PfCRT, the corresponding means were 85.4 (K76T) and

Table S1: *Docking enrichment and re-docking accuracy. Values are from corrected parent-study benchmarks using V2 grids. MMV Malaria Box values are consensus-score ROC-AUCs for PfDHFR, PfCRT, and PfATP4; DEKOIS 2.0 used AutoDock Vina for PfDHFR. BEDROC values use* $\alpha = 20$. *Re-docking RMSD is the heavy-atom RMSD between the top-scoring pose and the co-crystallized ligand; the MTX RMSD was approximately* 30 Å.

Target	Method	ROC-AUC	EF1 %	EF5 %	BEDROC	RMSD (Å)
PfDHFR (7F3Y)	Vina redock	–	–	–	–	N/A
	MMV consensus (Vina+DiffDock)	0.924	–	2.57	1.309	–
	DEKOIS (Vina only)	0.450	–	0.00	–	–
PfCRT (6UKJ)	MMV consensus (Vina+DiffDock)	0.971	–	1.11	9.434	–
PfATP4 (9N10)	MMV consensus (Vina+DiffDock)	1.000	–	2.00	1.810	–
PfClpR (4GM2; not PfClpP)	MMV enrichment	not evaluated	–	–	–	–

Target identity. PDB 4GM2 is PfClpR rather than PfClpP; it is not used as PfClpP validation.

Table S2: *Homology-model quality for the six resistance-associated states. Models were built with SWISS-MODEL from the indicated wild-type templates and retained only if QMEAN* > -4.0, *GMQE* > 0.7, *Ramachandran favored residues* $> 95\,\%$, *and Cα RMSD to the template* < 1.5 Å. *Detailed per-model scores are kept with the SWISS-MODEL outputs; the table records whether each model passed all four thresholds.*

Target	Mutant state	WT template (PDB)	QC gate
PfDHFR	N51I	7F3Y	PASS
PfDHFR	C59R	7F3Y	PASS
PfDHFR	S108N	7F3Y	PASS
PfDHFR	I164L	7F3Y	PASS
PfCRT	K76T	6UKJ	PASS
PfCRT	K76A	6UKJ	PASS

All six models were used as docking receptors. Passing these thresholds does not imply experimental structural accuracy.

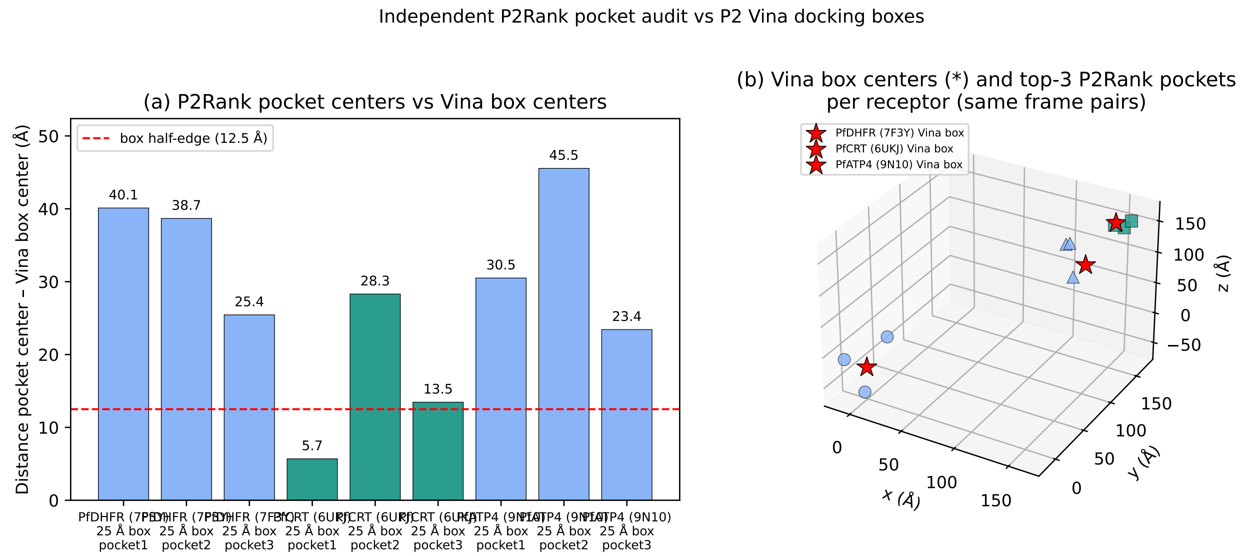

Figure S1: *P2Rank*[S1] *pocket centres versus the fixed Vina*[S2] *grid boxes. (a) Distance from the top three P2Rank pockets to the Vina box centre for PfDHFR (7F3Y), PfCRT (6UKJ), and PfATP4 (9N10); the dashed line is the* 12.5 Å *box half-edge. (b) Same centres in 3D (star: Vina box). For PfCRT the top pocket (score* 162.7*, probability* 0.999*) lies* 5.7 Å *from the box centre. Larger separations for PfDHFR and PfATP4 may reflect a different receptor frame in the parent-study grid files and are noted as provenance caveats, not as docking failures. P2Rank 2.5.1; exploratory check only.*

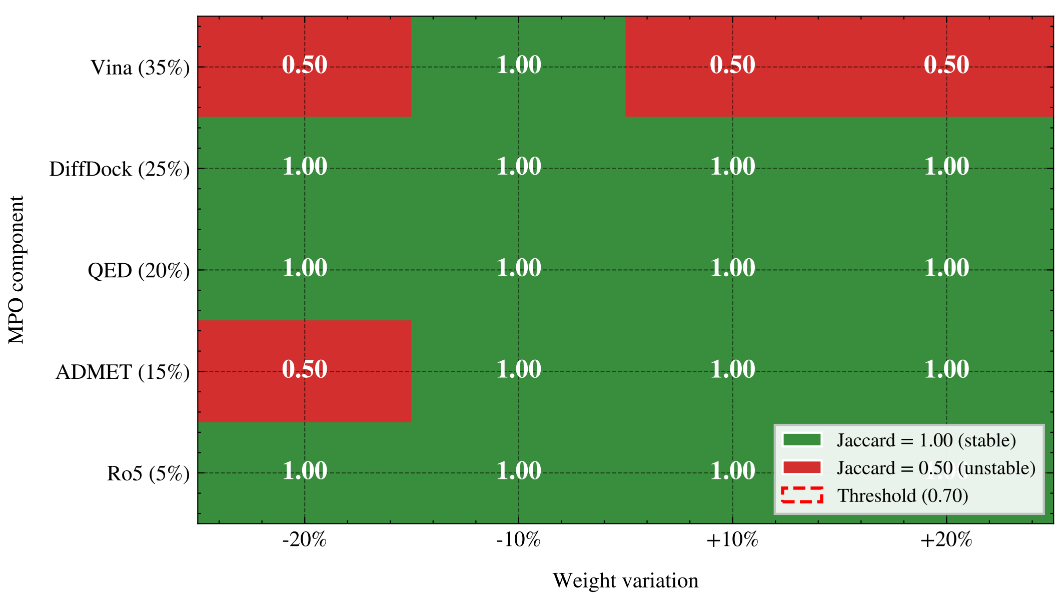

Figure S2: *MPO weight sensitivity. Each cell is the Jaccard similarity of the top-20 list after changing one MPO weight by* $\pm 20\,\%$*. Green (*1.00*) means no change; red (*0.50*) means substantial reordering.*

Table S3: *Predicted ADMET profile of the* 17 *set-C polypharmacology candidates (ADMET-AI, parent-library screening). All values are machine-learning predictions; experimental ADMET profiling and three-tool concordance were outside the scope of this study. Across candidates, LogS ranged from* 0.34 *to* 0.73*; CYP3A4 inhibition was* ≥ 0.5 *for* 7/17 *candidates, Caco-2 permeability was* ≥ 0.5 *for* 15/17*, and* $Clint_h$ *ranged from* 0.48 *to* 0.63.

Compound	ADMET-AI LogS	CYP3A4 prob.	Caco-2 prob.	Clint_h
PP-01	0.34	0.39	0.63	0.60
PP-02	0.46	0.97	0.86	0.55
PP-03	0.43	0.62	0.95	0.55
PP-04	0.40	0.58	0.92	0.63
PP-05	0.61	1.00	0.35	0.51
PP-06	0.54	0.90	0.97	0.60
PP-07	0.36	0.02	0.75	0.62
PP-08	0.59	1.00	0.41	0.56
PP-09	0.65	0.13	0.81	0.57
PP-10	0.49	0.13	0.93	0.62
PP-11	0.39	0.06	0.96	0.58
PP-12	0.49	0.00	0.98	0.48
PP-13	0.57	0.97	0.98	0.53
PP-14	0.66	0.05	0.68	0.49
PP-15	0.66	0.49	0.95	0.51
PP-16	0.54	0.25	0.91	0.50
PP-17	0.73	0.01	0.82	0.50

Table S4: *Candidate-cohort ACSI weight sensitivity. One ACSI component weight was varied by* $\pm 20\,\%$ *and the remaining weights were rescaled proportionally to preserve a unit sum. Mean and standard deviation refer to the resulting ACSI values across the* 17 *candidates. Rank stability is measured by Spearman correlation over all candidates and Jaccard overlap of the top five candidates with the baseline ranking. Because the full-library component matrix was unavailable, this analysis applies only to the 17-candidate Set-C cohort; it does not assess the full-library top-20 ranking.*

Weight varied	Variation (%)	New weight	Mean ACSI	SD	Spearman ρ	Top-5 Jaccard
D_{DrugBank}	-20	0.320	0.512	0.149	0.977 9	0.666 7
D_{DrugBank}	+20	0.480	0.574	0.159	0.933 8	0.666 7
D_{ANPDB}	-20	0.200	0.568	0.149	0.916 7	0.666 7
D_{ANPDB}	+20	0.300	0.518	0.158	0.980 4	0.666 7
f_{sp^3}	-20	0.160	0.552	0.152	0.951 0	0.666 7
f_{sp^3}	+20	0.240	0.534	0.153	0.958 3	0.666 7
NPL	-20	0.120	0.535	0.160	0.963 2	0.666 7
NPL	+20	0.180	0.551	0.145	0.980 4	1.000 0

Table S5: *Target-specific docking-score retention in the eligible candidates. Values are summarized across candidates with an eligible WT denominator for the indicated target. The target-specific class counts are descriptive and are not equivalent to the primary two-target classification.*

Target	Eligible candidates	Mean RRS	Median RRS	Range	Class A*/A	Class B/C/D
PfDHFR	12	75.1	72.0	58.9–131.8	2	10
PfCRT	17	86.2	84.0	70.4–110.2	10	7

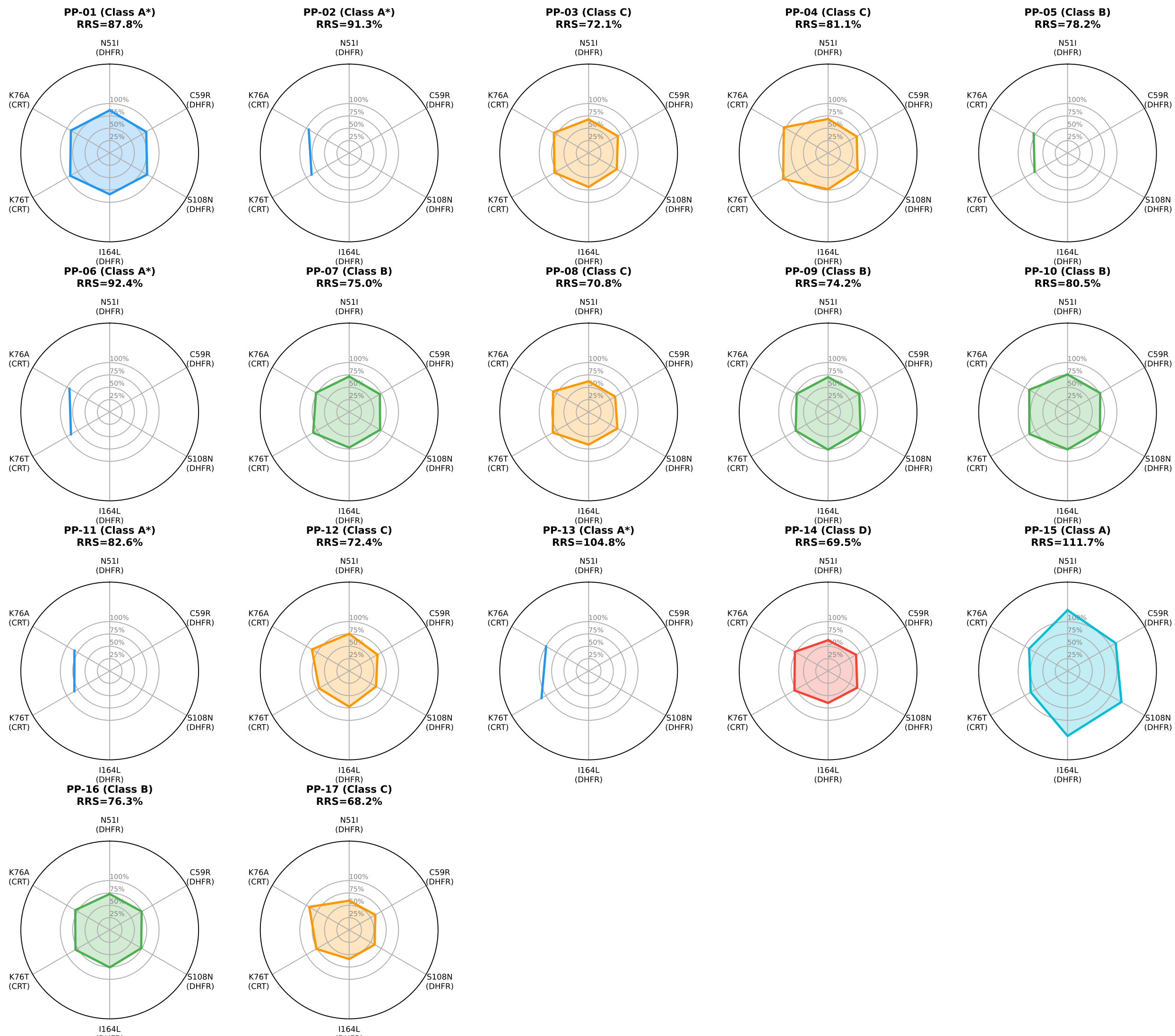

Figure S3: *Docking-derived RRS profiles for the 17 Set-C candidates across six resistance mutations. Missing axes mark an ineligible target under the* $5.0\,\mathrm{kcal\,mol^{-1}}$ *wild-type score threshold and are not treated as zero. Rings mark RRS of* $50\,\%$, $100\,\%$, *and* $150\,\%$*; the plot does not encode absolute binding strength.*

87.0 (K76A). These target-specific summaries describe score retention and do not measure biochemical resistance.

Table S6: *Sensitivity of the available-target RRS classes to the WT eligibility and retention thresholds. Counts are reconstructed from the versioned threshold-sensitivity output for all 17 candidates. Thresholds used in the main analysis are shown in bold. The table checks classification stability only; it does not calibrate cut-offs against experimental resistance.*

WT threshold	Lower retention	Upper retention	A*	A	B	C	D
5.0	70	80	**5**	**1**	**5**	**5**	**1**
5.0	75	85	3	1	4	3	6
5.0	80	90	2	0	4	2	9
4.0	70	80	5	1	5	5	1
6.0	70	80	6	0	5	5	1
7.0	70	80	6	0	5	5	1
7.0	75	85	4	0	4	3	6
7.0	80	90	2	0	4	2	9

Varying the wild-type eligibility threshold from $4.0\,\mathrm{kcal\,mol^{-1}}$ to $5.0\,\mathrm{kcal\,mol^{-1}}$ left class counts unchanged. Raising retention thresholds from 70 %/80 % to 80 %/90 % increased Class D from 1 to 9. These cut-offs are operational triage rules for this panel, not calibrated resistance categories.

Table S7: *Sensitivity of the PNS ranking to the imputed PfCRT centrality. The reference value is the observed network-mean imputation ($C_{\mathrm{PfCRT}} = 0.151$). Alternative values were zero, one-half, one-and-a-half, and twice the reference value. Spearman ρ is calculated against the reference 17-candidate ranking; all candidates have two target entries in the PNS record. The ranking is sensitive to this network-data choice; target essentiality and biological polypharmacology were not assessed.*

Imputation	C_{PfCRT}	Spearman ρ	Minimum PNS	Maximum PNS	n
Zero	0.0000	0.9716	0.300	5.147	17
One-half canonical	0.0755	0.9975	0.674	5.573	17
Canonical network mean	0.1510	1.0000	1.047	6.000	17
One-and-a-half canonical	0.2265	0.9755	1.421	6.426	17
Twice canonical	0.3020	0.9632	1.795	6.853	17

Secondary calibration and ranking analyses

Tartarus docking calibration

The Tartarus analysis used QuickVina[S3], an AutoDock Vina[S2] derivative, on 19 913 primary leads against PDB entries 1SYH, 6Y2F, and 4LDE. The composite MPO comparison included up to 17 211 complete observations. The composite association was $\rho = 0.0129$ ($p = 0.0913$); target-specific associations were small and do not imply score equivalence or independence.

Table S8: *Tartarus docking calibration: Spearman correlations between QuickVina scores and weighted MPO scores.*

Score	Spearman ρ	p-value	n	Interpretation
Composite (mean of 3 targets)	0.0129	9.13×10^{-2}	17 211	No meaningful association
1SYH (PfDHFR)	0.0655	8.18×10^{-18}	17 205	Small effect
6Y2F (PfCRT)	−0.0040	6.04×10^{-1}	17 211	Null
4LDE (A2A receptor)	−0.1415	1.17×10^{-77}	17 211	Small effect

Tartarus cross-docking correlation

Only 24 molecules overlapped between the Tartarus and project docking records. Because the structures were not matched, this comparison cannot calibrate the docking scores experimentally.

Table S9: *Tartarus independent-tool comparison for the 24-molecule overlap.*

Tartarus	Project metric	n	ρ	p	Interpretation
Composite	4GM2/PfClpR	24	0.246	2.47×10^{-1}	No
Composite	6UKJ/PfCRT	24	0.075	7.28×10^{-1}	No
Composite	7F3Y/PfDHFR	24	−0.110	6.10×10^{-1}	No
Composite	9N10/PfATP4	24	0.118	5.83×10^{-1}	No
1SYH/DHFR	7F3Y/PfDHFR	24	−0.442	3.04×10^{-2}	Exploratory

Complete ACSI components

ACSI was calculated as

$$\mathrm{ACSI} = 0.40 D_{\mathrm{DrugBank}} + 0.25 D_{\mathrm{ANPDB}} + 0.20 f_{sp^3} + 0.15\mathrm{NPL}.$$

Components were min–max normalized over the 17-candidate cohort. Two candidates exceeded ACSI > 0.70, and the cohort mean was 0.543. Varying one weight at a time by ±20 % gave Spearman correlations of 0.916 7–0.980 4 and top-five Jaccard overlaps of 0.666 7–1.000 0.

PNS ranking and imputation

The STRING[S4] network used confidence threshold 700; PfCRT centrality was imputed with the network mean, 0.151. The full ranking is listed in the table. Replacing the imputed value with zero, one-half, one-and-a-half, or twice the reference value gave rank correlations of 0.963 2–1.000 0. This result reflects sensitivity to a network-data choice, not biological target importance.

Table S10: *PNS ranking and RRS class. PfCRT-only denotes candidates without an eligible PfDHFR WT denominator for primary RRS.*

Candidate	PNS	ACSI	RRS class
Rank 1	6.00	0.599	C
Rank 2	4.71	0.308	C
Rank 3	4.53	0.567	C
Rank 4	4.48	0.603	B
Rank 5	4.48	0.551	B
Rank 6	4.45	0.173	B
Rank 7	4.45	0.459	A*
Rank 8	4.36	0.743	D
Rank 9	4.36	0.605	C
Rank 10	4.34	0.594	C
Rank 11	4.20	0.601	B
Rank 12	3.50	0.576	A
Rank 13	2.32	0.597	PfCRT-only
Rank 14	1.23	0.823	PfCRT-only
Rank 15	1.14	0.387	PfCRT-only
Rank 16	1.10	0.458	PfCRT-only
Rank 17	1.04	0.589	PfCRT-only

Predicted ADMET

The ADMET-AI[S7] endpoint table for all 17 candidates is given in Table S3. These predictions were not experimentally validated or compared across tools.

DiffDock–Vina and activity-model comparisons

In the 65 856-molecule library, Vina[S2] scores and DiffDock[S5] confidence were only weakly related overall. A Benjamini–Hochberg analysis of 18 pairwise activity comparisons across compound groups using three Ersilia[S6] models found no adjusted-significant differences; all rank-biserial effect sizes were negligible. These analyses provide no evidence of biological activity.

Table S11: *Questions addressed and interpretation boundaries. The Set-C docking cohort has 17 candidates and 136 wild-type/mutant systems after prioritization filters; 12 candidates have complete PfDHFR/PfCRT panels and define the primary RRS analysis, and 5 are PfCRT-only records reported separately. Parent-study MD systems do not overlap Set-C. The 16-system Set-C pilot passed geometric trajectory checks and is analyzed separately; its MD-RRS and MM-GBSA values are not used to assign docking-RRS classes. Docking-RRS is a computational triage estimand, whereas MD-RRS and MM-GBSA are secondary structural and endpoint diagnostics.*

Evidence layer	Cohort	Systems	Primary analyses	Interpretation boundary
Set-C docking	PP-01–PP-17	136 (12 complete; 5 PfCRT-only)	WT/mutant docking; target-specific RRS, ACSI, PNS	Primary computational triage cohort; not a measure of affinity, engagement, or biological resistance
Parent-study MD	Ligands 164, 201, 214, 438	4 WT complexes × 10 ns	RMSD, contacts, bound-state assessment, MM-GBSA where valid	Structural check; one interpretable MM-GBSA estimate (PfCRT–214)
Set-C MD pilot	PP-01/PP-02	16 systems (all met trajectory criteria)	Trajectory geometry, MD-RRS, MM-GBSA	Single-replicate structural stress test; non-equivalent to docking-RRS and separate from its classification
Full Set-C MD-RRS	PP-01–PP-17	136 planned systems	Full trajectory QC and MD-RRS	Not available because the complete cohort was not simulated

Identifiers were taken from PlasmoDB/VEuPathDB (accessed 12 August 2026)[S8]; the release/build string was not recorded. A parent-study PfCRT redocking check at pH 5.2 (100/100 library ligands; $\rho = 0.270$, $p = 0.006551$) was not merged into the Set-C RRS cohort. No GO or pathway enrichment was run, because a defined multi-gene targetome and parasite-proteome background were not available[S9].

Table S12: *Evidence scope and next validation. The computational results support prioritization and structural hypotheses within the stated cohorts; they do not substitute for biochemical, biophysical, phenotypic, or clinical measurements. The table separates the primary computational triage estimand from secondary diagnostics, technical checks, and unobserved biological quantities.*

Evidence layer	Supported interpretation	Next validation needed
Docking-RRS	Relative retention of empirical docking scores across the selected mutant panel; primary computational triage only	Matched biochemical activity measurements for wild-type and mutant PfDHFR/PfCRT
ACSI/PNS	Cohort-relative chemical-space and network-weighted ranking dimensions	Larger, independently selected panels and experimentally anchored target/network data
Parent-study MD	Ligand retention under the declared geometric criterion and system-specific endpoint assessment	Replicated, membrane-aware simulations and orthogonal binding or transport assays
Set-C MD pilot	Secondary trajectory-derived structural stress test for PP-01 and PP-02; single replicate per system and non-equivalent to docking-RRS	Independent mutant/wild-type replicates with declared convergence and uncertainty analyses
Biological polypharmacology	Not measured by the present computational scores	Target engagement plus parasite-phenotype and mechanism-of-action experiments

Table S13: *Scope of the targeted resistance panel. The panel was designed to test resistance-aware ranking across selected molecular contexts; it was not designed to estimate resistance prevalence, reconstruct the parasite targetome, or establish pathway-level polypharmacology.*

Dimension	Covered in this study	Not covered
PfDHFR	WT plus N51I, C59R, S108N, and I164L docking states	Other resistance haplotypes, prevalence, and experimental activity
PfCRT	WT plus K76T and K76A docking states; neutral-pH protocol	Full haplotype diversity, acidic-vacuole conditions, and transport measurements
PfATP4	One parent-study structural context	Mutant-state panel and resistance inference
PfClpR	One parent-study context associated with PDB 4GM2	PfClpP validation and broader proteostasis target coverage
Chemical cohort	17 filtered Set-C candidates; 12 define the complete two-target primary analysis	Representative sampling of the 65,856-molecule library
Biological outcome	Computational docking, trajectory, and endpoint summaries	Direct target engagement, parasite phenotype, and pathway-level mechanism

Table S14: *Stable Plasmodium falciparum target identifiers and mutation contexts. The identifiers define targets and states; they are not pathway-level or experimental evidence.*

Target	ID	Context	Mutation / boundary
PfDHFR	`PF3D7_0417200`	7F3Y WT template	N51I/C59R/S108N/I164L; antifolate states
PfCRT	`PF3D7_0709000`	6UKJ WT template	K76T/K76A resistance states
PfATP4	`PF3D7_1211900`	9N10 parent-study structure	Annotation only; outside Set-C docking panel
PfClpP	`PF3D7_0307400`	Proteolytic Clp subunit	Not assigned to 4GM2; no PfClpP claim
PfClpR	`PF3D7_1436800`	4GM2	4GM2 is PfClpR, not PfClpP

Table S15: *MM-GBSA for the four parent-study complexes. These systems are not among the 17 Set-C candidates. PfCRT–214 is reported as mean ± SEM over* 101 *snapshots; dissociated or conversion-invalid systems are N/A.*

Ligand	Target	ΔG_{bind}	SEM	Interpretation
164	PfClpR-labelled cohort*-WT	–	–	N/A; ligand dissociated‡
201	PfDHFR-WT	–	–	N/A; ligand dissociated‡
214	PfCRT-WT	−18.25	0.40	Interpretable (gmx_MMPBSA [S10])
438	PfATP4-WT	–	–	N/A; conversion corrupted§

*PDB 4GM2 is PfClpR, not PfClpP.
‡Ligand left the site during production; no binding free energy is inferred.
§Two-chain topology incompatible with CHARMM36 [S11]-to-AMBER conversion.

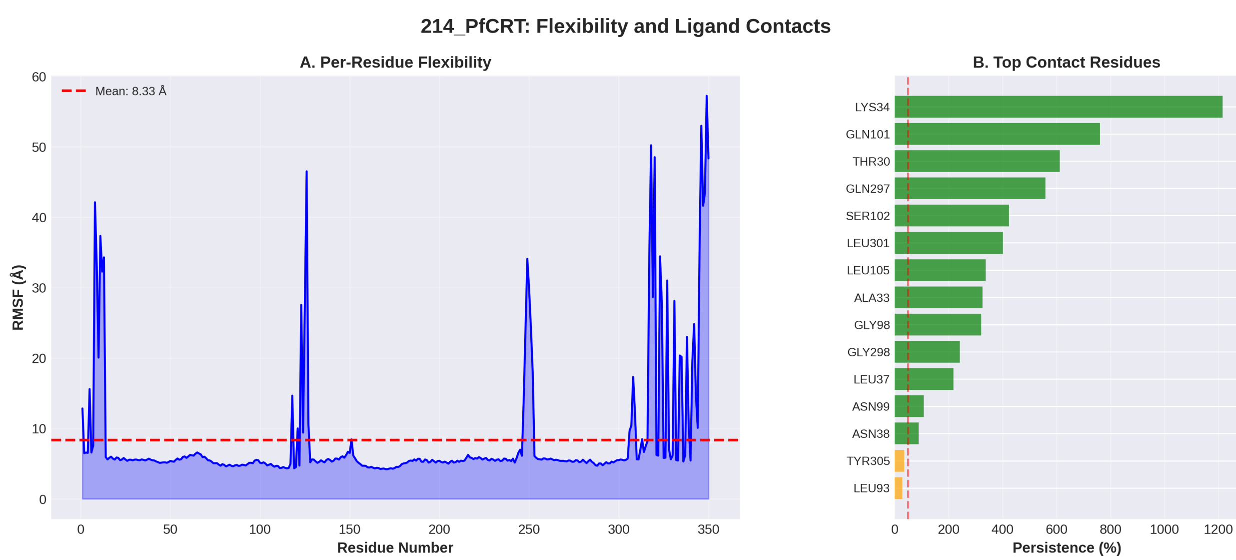

Figure S4: *Per-residue flexibility and contact persistence for the parent-study PfCRT–Ligand 214 complex. Lys34 is the main contact anchor; contact counts are aggregate, not fractional occupancy.*

The parent-study trajectories are a short, non-membrane explicit-solvent check on four wild-type complexes outside the Set-C list. Separately, a 16-system pilot ran PP-01 and PP-02 against the six-mutant panel for 10 ns each (single replicate per system). Trajectory-derived ratios and MM-GBSA are compared with docking RRS below. Neither design estimates long residence times or replicate uncertainty; per the JCIM MD-reporting guidelines[S12], at least three independent replicates per system are recommended. No multi-replicate campaign was launched for the presented estimates: the associated structural-stress extension on a single system (PP-01 PfDHFR WT, including a separate 25 ns trajectory; see below) is reported strictly as a secondary robustness check outside the Set-C estimand and is not a replicate campaign. Scope boundaries are summarized in Tables S12 and S13.

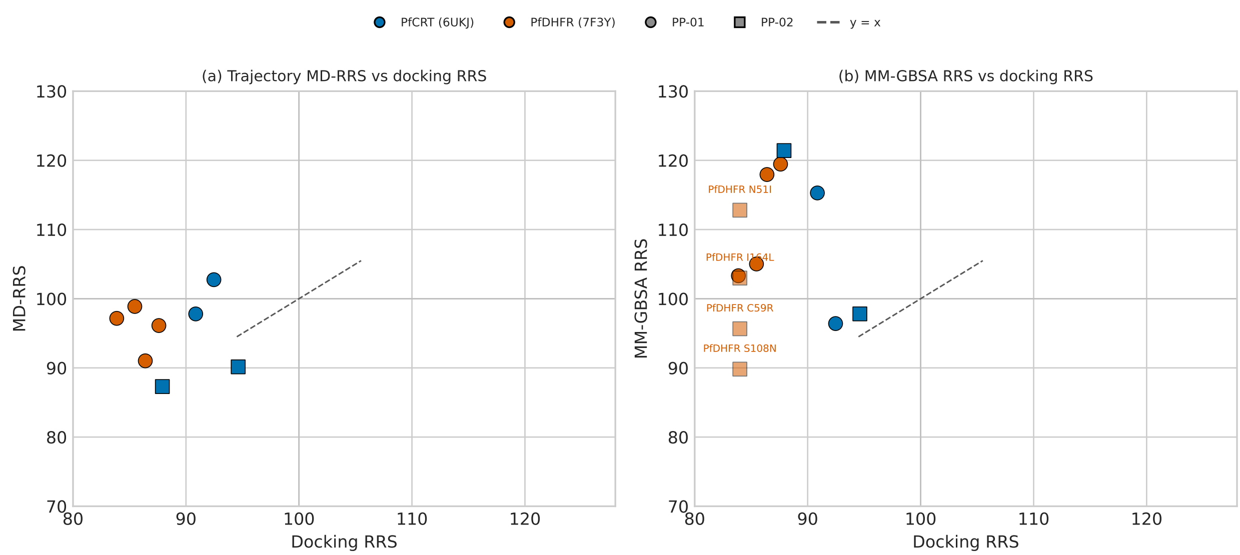

Figure S5: *Set-C pilot: trajectory-derived ratios versus docking RRS (wild type = 100). (a) Mean minimum-distance ratios for eight mutants with a docking wild-type reference. (b) MM-GBSA ratios versus docking RRS. Four PP-02 PfDHFR states lack a docking wild-type reference. The dashed diagonal indicates equality; the horizontal and vertical guides indicate the wild-type ratio. Docking predicted weaker mutant scores in the comparable systems; trajectory metrics did not recover that pattern. Neither quantity is a free-energy or experimental resistance measure.*

Convergence and interaction analysis

All nineteen MM–GBSA endpoints (GB$^{\mathrm{OBC2}}$ igb=5, 0.15 M salt; the sixteen Set-C pilot systems, the two PP-15 probe systems, and one independent PP-01 PfCRT WT replicate;

Table S16: *Set-C pilot MM-GBSA estimates.* ΔG_{bind} *is the mean over* 100 *snapshots;* SD_{prop} *is the propagated within-trajectory standard deviation, not a SEM or replicate uncertainty. MMG-RRS is the mutant/wild-type endpoint ratio (*$\times 100$*; wild type =* 100*). MD-RRS is the corresponding mean minimum-distance ratio. n/a: no docking wild-type reference for PP-02 PfDHFR. Two systems needed periodic-boundary reconstruction before endpoint analysis. Ratios above* 100 % *indicate retention within endpoint noise, not increased affinity.*

Candidate	Target	State	ΔG_{bind} (kcal mol^{-1})	SD_prop	MMG-RRS	MD-RRS	Docking RRS
PP-01	PfCRT	K76A	−35.29	1.17	115.3	97.8	90.86
PP-01	PfCRT	K76T	−29.51	2.65	96.4	102.7	92.47
PP-01	PfCRT	WT	−30.61	1.65	100.0	100.0	–
PP-01	PfDHFR	C59R	−26.56	1.91	105.0	98.9	85.47
PP-01	PfDHFR	I164L	−26.13	2.34	103.3	97.2	83.87
PP-01	PfDHFR	N51I	−29.83	1.50	118.0	91.0	86.40
PP-01	PfDHFR	S108N	−30.21	3.62	119.5	96.1	87.60
PP-01	PfDHFR	WT	−25.29	2.94	100.0	100.0	–
PP-02	PfCRT	K76A*	−24.53	1.27	97.8	90.1	94.62
PP-02	PfCRT	K76T	−30.45	1.38	121.4	87.3	87.91
PP-02	PfCRT	WT	−25.08	2.56	100.0	100.0	–
PP-02	PfDHFR	C59R	−28.85	1.88	95.7	102.0	n/a
PP-02	PfDHFR	I164L	−31.06	1.52	103.0	97.4	n/a
PP-02	PfDHFR	N51I	−34.02	2.68	112.8	72.2	n/a
PP-02	PfDHFR	S108N	−27.09	1.55	89.8	97.5	n/a
PP-02	PfDHFR	WT*	−30.16	2.06	100.0	100.0	–

Table S17: *Endpoint replicate consistency for the PP-01 PfCRT wild-type pillar. Two independent* 10 ns *productions executed under the identical MDP and analysis chain (*GB^{OBC2} *igb=5,* 0.15 M *salt,* 100 *snapshots) are compared. Tabulated spreads are the within-trajectory SD and the SEM over stored snapshots; the manuscript-wide quoted uncertainty elsewhere is the SEM. The SEM is informational here because MD snapshots are temporally autocorrelated. The between-replicate offset (*$|\Delta| = 2.01$ kcal mol^{-1}*) lies within one within-trajectory standard deviation of either run (inter-replicate mean* (-29.61 ± 1.42) kcal mol^{-1}*). Because the snapshots are temporally autocorrelated, the SEM does not quantify replicate-level uncertainty. The offset is therefore used as a conservative empirical noise floor of* 2.0 kcal mol^{-1} *for interpreting single-replicate mutant–wild-type endpoint contrasts.*

Replicate	Production	ΔG_{bind} (kcal mol^{-1})	SD	SEM
R1	production run 1	−30.61	4.12	0.41
R2	independent production run	−28.60	4.35	0.43

100 snapshots each except the PP-15 PfCRT probe at 80 snapshots, see Table S18) converge with a within-trajectory SEM below $1.0\,\mathrm{kcal\,mol^{-1}}$ (range $0.25 - -0.61\,\mathrm{kcal\,mol^{-1}}$; e.g. PP-01 PfCRT WT $(-30.61 \pm 0.41)\,\mathrm{kcal\,mol^{-1}}$). Because snapshots are temporally autocorrelated, this SEM characterises sampling within a single trajectory and does not capture inter-replicate variability. The empirical noise floor of $2.0\,\mathrm{kcal\,mol^{-1}}$ estimated from the two independent WT replicates (see the WT replicate-consistency table) is therefore retained for interpreting mutant–wild-type contrasts.

Extended structural stress test (PP-01 PfDHFR WT, 25 ns)

As an independent inter-trajectory robustness check on the pillar system of the primary cohort, a second PP-01 PfDHFR wild-type trajectory (fresh seed `26082901`, self-contained directory, the same OpenFF 2.2.0/CHARMM36m explicit-solvent protocol) was propagated to 25 ns under the identical trajectory-QC rule (heavy-atom protein–ligand minimum distance <5.0 Å for ≥10% of frames). The full 25.2 ns trajectory passes: bound fraction 1.000 and mean minimum distance 2.75 Å; the run was cleanly terminated by author decision at step 12 579 600 with a 25.16 ns recorded trajectory length and user-documented provenance rather than a natural end. This reproduces, over $2.5 \times$ the pilot window and on an independently initialised trajectory, the continuous bound state already reported for PP-01 PfDHFR WT in the 10 ns pilot (bound fraction 1.000, mean minimum distance 2.77 Å), showing that the pilot's fully-bound outcome is not an artefact of the short simulation window on this system. As with the pilot, no affinity, mutant contrast, or resistance conclusion is drawn from the extension: it is reported strictly as a secondary structural-stress test outside the Set-C estimand and does not constitute the multi-replicate campaign recommended by[S12].

Interaction fingerprints (ProLIF 2.2.1, 100 frames per trajectory, sixteen systems) show a sparse set of persistent contacts: four to eleven contacts per system exceed 50 % occupancy, most at 100 %. Recurring anchors include PfCRT TYR16 (fully occupied in five of six PfCRT systems) and PfDHFR LEU46/MET55, consistent with the canonical binding pockets. As a feasibility probe independent of the primary cohort, the single ligand PP-15 was docked to the wild-type states of PfCRT and PfDHFR (Vina 1.2.7) and prepared under the identical OpenFF 2.2.0/CHARMM36m explicit-solvent protocol used for the Set-C pilot. Both systems passed the pre-equilibration audits and completed EM–NVT–NPT equilibration and 10 ns production trajectories with valid checkpoints (Table S18); trajectory QC (heavy-atom protein–ligand minimum distance <5.0 Å for $\geq 10\%$ of frames) passes for both, with a bound fraction of 1.000 (mean minimum distances 2.77 Å PfDHFR and 3.20 Å PfCRT). MM-GBSA estimates on PBC-whole trajectories ($\mathrm{GB}^{\mathrm{OBC2}}$ igb=5, 0.15 M salt) are $(-29.52 \pm 0.33)\,\mathrm{kcal\,mol^{-1}}$ (PfDHFR) and $(-27.79 \pm 0.49)\,\mathrm{kcal\,mol^{-1}}$ (PfCRT), and are reported in Table S18. The probe is outside the 17-candidate Set-C estimand: no mutant or RRS estimate is reported for PP-15, and the single-replicate endpoints serve only as within-protocol feasibility contrasts. Docking-mode reproducibility for PP-15 was checked across five random seeds on the same grid box used for its canonical single-run score (Table S21): the five-seed dispersion is $0.018\,\mathrm{kcal\,mol^{-1}}$ (PfDHFR) and $0.030\,\mathrm{kcal\,mol^{-1}}$ (PfCRT), i.e. $\leq 0.03\,\mathrm{kcal\,mol^{-1}}$ on both receptors. The canonical PfCRT score ($-7.810\,\mathrm{kcal\,mol^{-1}}$) lies at the edge of the seed range; the canonical PfDHFR score ($-8.621\,\mathrm{kcal\,mol^{-1}}$) is $0.03\,\mathrm{kcal\,mol^{-1}}$ stronger than the seed mean ($-8.590\,\mathrm{kcal\,mol^{-1}}$), i.e. outside the seed range but within the same seed-noise floor. The seed choice therefore does not drive the probe ranking. The same control was repeated for PP-01 on the canonical inputs of its wild-type scores (RRS-pilot ligand preparation and the P1-V2 wild-type grid boxes; Table S21): mode-1 affinity is stable to within $0.05\,\mathrm{kcal\,mol^{-1}}$ on both wild-type receptors, and the canonical PfDHFR WT score ($-7.50\,\mathrm{kcal\,mol^{-1}}$) falls inside the five-seed spread (mean $(-7.499 \pm 0.021)\,\mathrm{kcal\,mol^{-1}}$). For PfCRT WT the canonical value ($-9.30\,\mathrm{kcal\,mol^{-1}}$) lies $0.04\,\mathrm{kcal\,mol^{-1}}$ above the observed range ($-9.262 - - -9.225\,\mathrm{kcal\,mol^{-1}}$) because the original receptor-preparation detail of that single run is not preserved on disk; this is within the $0.05\,\mathrm{kcal\,mol^{-1}}$ seed-noise floor, so the seed-sensitivity conclusion is unchanged. Within the reproduced ligand, receptor, and grid configuration, the tested seed choices produced a dispersion no greater than $0.05\,\mathrm{kcal\,mol^{-1}}$.

This supports protocol-local seed stability, but does not establish exact reproducibility of the historical PfCRT WT score because the original receptor preparation was not archived.

Table S18: *PP-15 single-ligand feasibility probe (outside the Set-C estimand). Vina scores are best-pose affinities on wild-type receptors; MD columns report the OpenFF 2.2.0/CHARMM36m protocol outcome for the canonical* 10 ns *production runs (self-contained replicate directories). Trajectory QC passes for both systems (bound fraction* 1.000*; mean minimum distances* 2.77 Å *PfDHFR and* 3.20 Å *PfCRT). MM-GBSA is* GB^{OBC2} *(igb=5),* 0.15 M *salt, mean over the frame window* $\pm$ *SEM (*100 *frames for PfDHFR,* 80 *for PfCRT; PBC-whole trajectories). No mutant or RRS estimate is reported.*

Target	Vina (kcal mol^{-1})	Audit	Production	MM-GBSA (kcal mol^{-1})	SEM	Bound fracti
PfDHFR (7F3Y)	−8.62	PASS	10 ns complete	−29.52	0.33	1.000
PfCRT (6UKJ)	−7.81	PASS	10 ns complete	−27.79	0.49	1.000

The versioned machine-readable outputs are `results/v2609_rrs_margin_20260830/rrs_threshold_marg` `rrs_threshold_margins_by_candidate.csv`, and `margin_analysis_manifest.json`. The input SHA-256 and analysis parameters are recorded in the manifest.

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Use of Artificial Intelligence

AI assistants were used to help with code and analysis scripts. All four Springer Nature AI-policy expectations were respected: (i) human accountability for every scientific claim remains with the listed authors; (ii) AI supported but did not replace scholarly judgement, editorial decision-making or the design of the comparison; (iii) the use of AI is disclosed here transparently; (iv) no confidential manuscript material, peer-review content or restricted data were shared with external AI services. All AI-assisted text and code were reviewed and edited by the authors. No generative AI tool is an author. Computational results, statistics, and quantitative claims were produced and checked by the authors.

Table S19: *Lightweight robustness and transfer audits for the selected Set-C cohort. These post-selection diagnostics assess sensitivity of the computational ranking and class definitions; they do not establish biochemical affinity, biological resistance, target engagement, or mechanism of action.*

Audit	Result	Interpretation boundary
Leave-one-candidate-out	Complete two-target $n = 12$; PNS–RRS ρ ranged from -0.3182 to 0.0273; ACSI–RRS ρ ranged from -0.7636 to -0.3182	Rank associations are sensitive to candidate removal and remain exploratory
Target-stratified correlations	PfDHFR: PNS–RRS $\rho = -0.3497$; PfCRT: $\rho = -0.0417$	Target-specific patterns are not interchangeable with the pooled analysis
Score perturbation	$\pm 1\,\mathrm{kcal\,mol^{-1}}$ perturbations generated 272 candidate–score records; class unchanged in 212/272 (77.9 %)	Operational RRS classes show threshold sensitivity, especially to WT-score perturbation
Threshold sensitivity	The threshold grid used in the main analysis was retained unchanged in the source record	Thresholds are triage rules, not experimentally calibrated categories
Related ChEMBL transfer	The public ChEMBL panel used by the companion study was verified against its source record; no P2-specific external panel was available	Related-project transfer evidence is not an independent P2 replication or experimental validation
External docking replication	39 ligands, 8 PfDHFR/PfCRT states, 312/312 finite records after declared EXT-039 embedding exclusion; bootstrap mean RRS 100.45 (99.59–101.32), Class-A fraction 0.974 (0.923–1.000)	Computational docking sensitivity analysis only; not experimental validation or an MD estimate
GNINA CNN consensus rescoring	GNINA 1.3.2 CNN affinity rescored the same 312 poses (post-processing only); 312/312 finite scores. Under the frozen RRS estimand (target eligibility decided on the canonical Vina scale), 38/38 eligible ligands were class A under both scoring layers, 0 class discordances; mean RRS Vina 100.4 vs CNN 103.6; Spearman (ligand mean RRS) 0.558; per-mutant ρ: K76T 0.644, K76A 0.486, N51I 0.271, S108N 0.204, I164L 0.040, C59R -0.076	Consensus sensitivity: class-level retention is shared across scoring functions; per-mutant rank transfer is not supported (Vina-only remains canonical)
Set-C bootstrap CI95	$B = 1 \times 10^4$ resamples (seed 42) on the 17-candidate class table: class fractions A* 0.294 [0.118–0.529], A 0.059 [0.000–0.176], B 0.294 [0.118–0.529], C 0.294 [0.118–0.529], D 0.059 [0.000–0.176]. Per-mutant retention (WT = 100): PfCRT K76T 85.4 [81.0–90.4], K76A 87.0 [83.1–90.9] ($n = 17$, intervals exclude 100); PfDHFR N51I 74.7 [67.2–85.2], C59R 73.7 [67.4–82.1], S108N 75.3 [67.7–85.8], I164L 76.8 [68.9–88.2] ($n = 12$, intervals overlap 100 for I164L)	PfCRT retention-not-gain is robust to resampling; PfDHFR intervals are wide and do not support per-mutant rank claims
MD-filter retention gate (pilot scope)	Intersection of docking-RRS $\geq$ 80 % with pilot MD-RRS $<$ 100 on the 12 gate-eligible rows (PP-01/PP-02): 7/12 rows pass both gates. PP-01 satisfies the two-target gate (PfCRT via K76A 97.8; PfDHFR four of four mutants 91.0–98.9); PP-02 does not (PfDHFR rows lack a docking wild-type anchor; single-target gate). PP-01 PfCRT K76T (102.7) is the sole row above parity and lies within the $2.0\,\mathrm{kcal\,mol^{-1}}$ noise floor	Pilot-scope consistency check; full-panel (17 × 8) MD-RRS is `NOT_COMPUTED`; the gate does not replace docking-RRS and is not a biological resistance claim

Table S20: *Use of companion P1 evidence in P2. P1 results are included only as methodological or chemical-space context. They do not validate P2 mutant-state RRS, biological resistance, or polypharmacology.*

P1 evidence	Permitted use in P2	Boundary
Redocking and enrichment benchmarks	Context for pose reproduction and general ligand-ranking behavior of the docking workflow	Does not validate mutant-state score retention or RRS
Physicochemical and drug-likeness profiles	Context for describing the upstream chemical library and the selected PP-01–PP-17 cohort; the cohort audit found 17/17 exact SMILES matches	Shared candidate identity and provenance preclude independent-replication claims; properties are not biological activity measurements
Chemical-space and scaffold analyses	Context for the origin and diversity of the parent library	Set-C remains a filtered prioritization cohort, not a representative sample
MPO sensitivity	Context for the upstream selection procedure and its local stability	Does not test RRS or network-weighted ranking
P1–P2 shared candidates or provenance	Used only with explicit cohort and provenance labeling	Not an independent replication when candidates or source workflows overlap

Table S21: *Multi-seed redocking reproducibility for candidates PP-15 and PP-01 (AutoDock Vina[S2], five random seeds; lowest-energy mode). Values in* $\mathrm{kcal\,mol^{-1}}$. *Each control reuses the exact canonical ligand preparation and grid box that produced the candidate's canonical single-run score. For PP-01 PfCRT WT, the canonical value (*-9.300*) lies* $0.04\,\mathrm{kcal\,mol^{-1}}$ *above the observed seed range because the original receptor-preparation detail of that single run is not preserved on disk; this is within the* $\leq 0.05\,\mathrm{kcal\,mol^{-1}}$ *seed-noise floor. Seed-sensitivity controls measure docking-stochasticity reproducibility, not binding or resistance.*

Candidate–Target	Mean	Range	Canonical run	Seed sensitivity
PP-15 PfDHFR WT	-8.590 ± 0.007	$-8.603/-8.585$	-8.621	$\leq 0.03\,\mathrm{kcal\,mol^{-1}}$
PP-15 PfCRT WT	-7.822 ± 0.011	$-7.840/-7.810$	-7.810	$\leq 0.03\,\mathrm{kcal\,mol^{-1}}$
PP-01 PfDHFR WT	-7.499 ± 0.021	$-7.529/-7.475$	-7.500	$\leq 0.05\,\mathrm{kcal\,mol^{-1}}$
PP-01 PfCRT WT	-9.250 ± 0.015	$-9.262/-9.225$	-9.300	$\leq 0.05\,\mathrm{kcal\,mol^{-1}}$

Table S22: *Descriptive margins of the available-target RRS observations around the operational 70% and 80% thresholds. A candidate is flagged when at least one finite observation lies within 5 percentage points of either threshold. The analysis used the frozen canonical classification table and did not modify class assignments; it is sensitivity information rather than calibration against experimental resistance.*

Candidate	Finite RRS observations	Minimum absolute margin (percentage points)	Near-threshold observations	Flagged
PP-01	12	3.87	2	Yes
PP-02	4	7.91	0	No
PP-03	12	0.00	12	Yes
PP-04	12	1.25	8	Yes
PP-05	4	0.67	4	Yes
PP-06	4	10.61	0	No
PP-07	12	1.84	12	Yes
PP-08	12	2.76	8	Yes
PP-09	12	0.56	12	Yes
PP-10	12	3.64	8	Yes
PP-11	4	2.50	4	Yes
PP-12	12	0.38	8	Yes
PP-13	4	19.40	0	No
PP-14	12	1.20	8	Yes
PP-15	12	6.24	0	No
PP-16	12	0.11	12	Yes
PP-17	12	3.24	2	Yes

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